IDS Project Report

Swarm Intelligence and Target Search using AoA and Voronoi

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Compiled on March 13, 2026

Abstract

A "treasure hunt" can be a fun and engaging opportunity to test the capabilities of a swarm of robots in a simulated environment. The multiagent system proposed in this project is designed to search for a target in a 2D environment using a combination of swarm intelligence, Voronoi cells, Lévy flight, Angle of Arrival (AoA), and sensor fusion techniques. The main objective of the project is to evaluate the performance of the proposed multiagent system in terms of search efficiency, convergence time, and robustness to noise and uncertainties in the environment. The simulation will be implemented in Python, and the results will be analyzed to draw conclusions about the effectiveness of the proposed approach in solving the treasure hunt problem.

Keywords: *Swarm Intelligence, Voronoi Cells, Lévy Flight, Angle of Arrival, Sensor Fusion*

This project presents a control framework for a swarm of mobile robots that find, estimates and aligns around a target. The system combines a lidar, for obstacle detection, an Extended Kalman Filter (EKF) for state estimation and linear consensus for coordination. The robots are able to estimate the position of the target using Angle of Arrival (known also as Bearing angles). The detection of the target is performed by a simple thresholding of the received signal strength emitted by the treasure, which is assumed to be a weak radio emitter. The robots are able to coordinate their movements using a linear consensus algorithm, which allows them to converge to the estimated position of the target while avoiding obstacles. The performance of the proposed control framework is evaluated through simulations in a 2D environment, where the robots are tasked with finding and aligning around a target while navigating through a cluttered environment. The results show that the proposed approach is effective in enabling the swarm of robots to find and align around the target, even in the presence of noise and uncertainties in the environment.